Alka Madan

Senior Edge Security Architect · DDoS & WAAP / WAF · Akamai · Cloudflare · Imperva · Security Migration & Cost Optimization

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PROFESSIONAL SUMMARY

Senior Edge Security Architect with 7+ years of experience leading DDoS and WAAP/WAF protection programmes across large-scale enterprise and global multi-entity environments. Proven track record driving security migration projects — from planning and policy validation through to post-deployment support — across CDN platforms including Akamai, Cloudflare, and Imperva. Deep expertise assessing security posture, identifying gaps, and implementing improvements aligned with global cybersecurity standards. Experienced in raising and managing service requests and incident reports with vendors, delivering KPI-driven security dashboards, and communicating complex security findings to stakeholders across multicultural, cross-functional teams. Hands-on with log analysis, traffic forensics, and threat mitigation across L3/L4/L7. Team lead and mentor who translates technical edge architecture into measurable business outcomes including $200M+ in cumulative revenue protection and direct origin cost reduction.

CORE SKILLS

CDN & Edge Platforms

· Akamai (Expert — 7 yrs)
· Cloudflare (WAF, DDoS, migration)
· Imperva (migration, policy design)
· WAAP / App & API Protector (KONA, ASE)
· Bot Manager Premier (BMP SME)
· Cloudlets / Dynamic Origin

Security Disciplines

· DDoS — L3/L4/L7 mitigation
· API Security & abuse prevention
· OWASP: XSS, SQLi, RFI, CMDi
· Network segmentation & ACLs
· TCP/IP · TLS · DNS · WireShark
· CDN offload & origin cost reduction

Cloud & Infrastructure

· Azure (NSGs, IP allow listing)
· Terraform IaC · Python & Bash
· Linux (Ubuntu — networking)
· Nginx · SRE (incident response)
· mPulse / RUM / CrUX
· KPI dashboards & stakeholder reporting

PROFESSIONAL EXPERIENCE

GRANICUS *2024 — Present*

Senior Edge Architect / Senior DevSecOps Engineer

- Design, engineer, and operate distributed edge-based DDoS and WAAP/WAF protection systems securing multiple production SaaS applications in a high-availability 24/7 environment; continuously assess security posture, identify gaps, and drive improvements aligned with enterprise cybersecurity standards.
- Architect layered security controls combining CDN filtering, WAF policies, IP-based ACL segmentation, and Azure NSG ingress restrictions.
- **Led zero-downtime migrations from Imperva and Cloudflare to Akamai** across live traffic environments, translating policy logic and WAF rule sets between platforms with no service disruption.
- Reduced edge infrastructure costs by migrating from Akamai SiteShield to strict IP ACL-based security zones, eliminating the SiteShield contract cost while strengthening origin hardening and reducing exposed attack surface.
- Drive continuous improvement of security policies through Terraform automation; authored operational runbooks, incident reports, and KPI-driven security dashboards to communicate security posture to stakeholders across SRE and platform teams.
- **Mentor junior and mid-level engineers** on traffic debugging, packet-level reasoning, and security incident triage.

EXPEDIA GROUP *2023 — 2024*

Security Engineer 3

- **Led a 3-person edge security team**, coordinating day-to-day security operations, incident triage, and WAF governance across Expedia's global travel platform.

- Fine-tuned ~30 Akamai WAAP/WAF and DDoS security configurations, reviewing and validating security policies to improve platform resilience — contributed to **~$60M USD in revenue protection**.
- Drove API security and WAAP/bot management initiatives including vendor evaluations, proofs of concept, and onboarding coordination; raised and tracked service requests and incident tickets with Akamai to ensure timely resolution.
- Reduced false positives by analyzing traffic patterns and tuning WAF and bot detection rules; supported enterprise Zscaler integrations aligning edge protections with secure access flows.

AKAMAI TECHNOLOGIES *2019 — 2023*

Security Architect — Bot Management SME

- Served as internal SME for Akamai WAAP (Bot Manager Premier and KONA WAF), supporting enterprise customer escalations, onboarding new entities onto security platforms, and leading complex multi-layer DDoS and application security investigations.
- Protected enterprise applications from L3/L4/L7 attacks including application-layer DDoS, credential stuffing, scraping, and OWASP-class abuse patterns.
- Built and deployed custom WAF signatures using regex and XML to address emerging threats and zero-day-style attack payloads; performed deep false-positive/negative analysis improving detection accuracy while preserving legitimate traffic.
- Advised enterprise customers on edge-based WAAP security architecture; produced HLD/LLD design documents, onboarding process documentation, and security policy review reports for stakeholders across diverse global teams.
- Delivered training to customers, partners, and internal teams on web application security, traffic behavior, and defense strategies.

Solutions Engineer

- Troubleshot CDN and edge network issues spanning caching behaviour, TLS handshakes, routing, and origin performance across distributed enterprise environments.
- Performed log-based forensics and traffic analysis using WireShark to identify client–edge–origin communication failures; investigated latency spikes, packet anomalies, and routing inconsistencies.
- Supported Akamai services including CDN, GTM, Cloudlets, WAF, Bot Manager, and DataStream across enterprise accounts.

INDEPENDENT CONSULTING

GLOBAL HOSPITALITY & TRAVEL CLIENT *2025*

Akamai Security Consultant

- Conducting full-stack Akamai property audit across two production properties serving 6B+ edge hits/month, covering CDN offload optimization, WAF and Bot Manager governance, Core Web Vitals, and mPulse observability.
- Architected a blue-green deployment mechanism at the edge using Akamai variables and Dynamic Origin Hostname generation, enabling the engineering team to redirect live traffic to sandbox review environments without infrastructure changes or DNS involvement.
- Driving CDN offload improvement from 87% toward 95%+ by engineering cache rules for low-offload content types (JSON at 51%, MP4 at 73%, /is/content/ objects at near-zero offload) directly reducing origin compute and bandwidth costs at 6B+ hit/month scale and eliminating ~13.8M wasted origin round trips/month.
- Leading full strategic revamp of underutilized Akamai Bot Manager Premier deployment — rebuilding detection logic, policy segmentation, and traffic classification from the ground up to maximize ROI on the contract.
- Implementing Akamai Account Protector for credential stuffing and ATO protection; managing Cloudlet policies for Edge Redirect and request-control logic.

GLOBAL CONSUMER GOODS CLIENT (E-COMMERCE) *2024*

Akamai WAF & Bot Management Consultant

- Delivered end-to-end Akamai WAF and Bot Manager consultancy for a global consumer goods e-commerce platform, including policy design, rule tuning, and false-positive reduction across production traffic.
- Managed Cloudlet policies to implement edge-layer redirect and request-control logic; strengthened origin hardening ensuring only sanitised, validated traffic reached backend systems during peak e-commerce events.

EDUCATION

Bachelor of Engineering *2019*

Visvesvaraya Technological University · CGPA: 9.2 / 10